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The Factor of Scale in Remote Sensing

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Thanks to such second- and third-generation sensor systems as Thematic Mapper, SPOT, and AVHRR, a user of digital satellite imagery for remote sensing of the earth's surface now has a choice of image scales ranging from 10 m to 1 km. The choice of an appropriate scale, or spatial resolution, for a particular application depends on several factors. These include the information desired about the ground scene, the analysis methods to be used to extract the information, and the spatial structure of the scene itself. A graph showing how the local variance of a digital image for a scene changes as the resolution-cell size changes can help in selecting an appropriate image scale. Such graphs are obtained by imaging the scene at fine resolution and then collapsing the image to successively coarser resolutions while calculating a measure of local variance. The local variance/resolution graphs for the forested, agricultural, and urban/suburban environments examined in this paper reveal the spatial structure of each type of scene, which is a function of the sizes and spatial relationships of the objects the scene contains. At the spatial resolutions of SPOT and Thematic Mapper imagery, local image variance is relatively high for forested and urban/suburban environments, suggesting that information-extracting techniques utilizing texture, context, and mixture modeling are appropriate for these sensor systems. In agricultural environments, local variance is low, and the more traditional classifiers are appropriate.

Introduction

One of the fundamental characteristics of a remotely-sensed image is spatial resolution, or the size of the area on the ground from which the measurements that comprise the image are derived (Townshend, 1980). In this sense, spatial resolution is analogous to the scale of the observations. In most scientific endeavors the investigator selects the scale at which observations are collected. However, when using remotely-sensed imagery from space-borne sensors, investigators are limited to specific scales of observations. Until recently, this choice was extremely limited with data from the Landsat Multispectral Scanner (MSS) (80-m resolution) being the most commonly used.

However, new satellite platforms have been launched with sensors at several new spatial resolutions. A few notable examples include the Thematic Mapper (TM) with 30-m resolution which was included on the fourth and fifth Landsat satellites (Engel and Weinstein, 1983); SPOT, a French satellite has a 10-m panchromatic band and 20-m multispectral capabilities (Midan et al., 1982); and the MESSR sensor on the planned Japanese Marine Observation Satellite which will have 50-m resolution (Ishizawa, 1981). At the same time that data with finer spatial resolutions are becoming available, data with more coarse resolution such as from the AVHRR (1.1- and 4-km resolution) are increasingly being used (Tucker et al., 1985).

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Associated with this enhanced variety of data types is the burden of choice. While there are considerations beyond spatial resolution concerning the spectral, temporal, and radiometric characteristics of data, it is becoming apparent that the factor of scale plays an increasingly important role in the planning of remote sensing investigations. In the past, the selection of an appropriate scale has been left to the experience and intuition of individual investigators. A method is presented in this paper that provides information that can help in the selection of appropriate image scale.

The appropriate scale for observations is a function of the type of environment and the kind of information desired. The techniques used to extract information from imagery also interact with these variables to influence the selection of an appropriate scale. Thus, the problem of selecting an appropriate scale is fairly complex. The possible combinations of scales (or spatial resolution), analysis methods, environments, and questions about those environments is essentially infinite, making the enumeration of all combinations an enormous task. Therefore, a method is needed that helps individual investigators, already familiar with the environment and the information they desire about it, to select an appropriate combination of spatial resolution and analysis methods. The approach proposed in this paper is based on the spatial structure of images, which is an indication of the relationship between environments and spatial resolution. A new method of measuring the spatial structure of images as a function of spatial resolution is presented. This method graphs local variance in images as a function of their spatial resolution. Combined with an understanding of the assumptions of various types of analysis methods, these graphs can help investigators select an appropriate combination of spatial resolution and analysis methods.

Throughout this paper it is necessary to discuss the characteristics of environments in a generic manner. In order to do this, a model for the nature of scenes is required (Strahler et al., 1986). The model employed in this paper assumes scenes are composed of *discrete objects*. These objects are arranged either in a *mosaic* that completely cover the area, or distributed on a continuous *background*. In this model, *elements* are abstractions of real objects in the ground scene. The elements in a scene model may vary, depending on the interests of the investigator. For example, in a forest scene, elements might be individual leaves, branches, trees, or stands of trees. Common backgrounds in a forest scene include soil, snow, and vegetative understory. Models of scenes can be either simple or complex. *Simple* scene models contain only one class of element and the background, while *complex* scene models are those which include more than one class of element. A residential scene is an example of a complex scene best characterized as a mosaic of elements. The elements in the scene model include houses, lawns, streets, cars, trees, and gardens. They are best described as being fit together in a mosaic, as opposed to being arranged over a common background.

Nested models of scenes are also possible, in which one level of elements is used to derive the properties of a new level, or larger elements. Again, forest scenes serve as a good example. At one level, the individual trees might serve as elements

in a scene model. At the same time, groups of trees, or stands, might serve as a higher level in the scene model. In this situation, the properties of the individual trees are used to determine the characteristics of the stands. This method of describing scenes using models is developed more fully in a previous paper, to which interested readers are referred for greater detail (Strahler et al., 1986).

Methods

To select an appropriate scale of data, one must understand the spatial structure of images. In particular, the manner in which images of a scene change as a function of spatial resolution is important. Several authors have investigated the spatial structure of images, usually at one or two discrete resolutions. Craig and Labovitz (1980) measured autocorrelation in Landsat MSS images and tested the influence of factors related to the sensor, physical factors such as sun angle and cloud cover, and a "geographic location" factor. Labovitz et al. (1980) extended the study to include images at the resolution of the Landsat Thematic Mapper (30 m) and found autocorrelation to be higher in images with finer spatial resolution. Woodcock and Strahler (1985) used one- and two-dimensional variograms to investigate the spatial structure of both simulated and real images. Simonett and Coiner (1971) more directly addressed the effect of spatial resolution on image structure by overlaying grids on aerial photographs and counting the number of land-use categories that occurred in each cell. By using different size grids, the effect of changing spatial resolution was evaluated. They demonstrated that the number of pixels containing more than one land-cover type was a function of both the complexity of the scene and the spatial resolution of the sensor.

The spatial structure of images is expected to be primarily related to the relationship between the size of the objects in the scene and spatial resolution. Graphs of local variance in images as a function of spatial resolution may be used to measure spatial structure in images. The reasoning behind this measure is as follows. If the spatial resolution is considerably finer than the objects in the scene, most of the measurements in the image will be highly correlated with their neighbors and a measure of local variance will be low. If the objects approximate the size of the resolution cells, then the likelihood of neighbors being similar decreases and the local variance rises. As the size of the resolution cells increase and many objects are found in a single resolution cell, the local variance decreases.

Calculation of the data for these graphs is accomplished by degrading the image to successively more coarse spatial resolutions, while measuring local variance at each resolution. Local variance is measured as the mean value of the standard deviation of a moving 3×3 window. In an image, each pixel (except around the edge) can be considered the center of a 3×3 window. The standard deviation of the nine values is computed, and the mean of these values over the entire image is taken as an indication of the local variability in the image.

To measure local variance at multiple resolutions, the image data are degraded to coarser spatial resolutions. The algorithm used to degrade the imagery simply averages resolution cells to be combined into a single, larger resolution cell. This approach implies an idealized

square-wave response on the part of the sensor, or that the measurement produced by the sensor is derived only from the area inside the pixel. This assumption is unrealistic, however; the intent is to study the basic relationships involved, not to mimic the response of a particular sensor. This approach avoids the controversies over the exact definition of spatial resolution and how it should be measured (Townshend, 1980).

One effect of the degradation of images is that the number of pixels decreases as resolution becomes more coarse. Thus, there are a limited number of times that an image can be degraded and still have a reasonable number of pixels to estimate local variance. When possible, large images were used for calculation of the data for these graphs. As a rule of thumb, the minimum size of images used to measure local variance was about 60 pixels on a side.

The graph of local variance as a function of resolution is similar in intent to techniques used in geography and ecology to find the "scales of action" (Mollering and Tobler, 1972; Greig-Smith, 1964). These techniques are similar to spectral analysis and designed to determine the scales at which the major portion of processes occur (Tukey, 1961; Rayner, 1971). The local variance measure is similar to the covariance portion of traditional measures of autocorrelation for short distances (Cliff and Ord, 1981). There are, however, certain limitations involved in the use of graphs of local variance as a function of resolution. One limitation is that it is used on a single image and does not take into account multispectral variation. This limitation is not serious as it could be generalized to the multidimensional case. A more serious limitation is that it is dependent on the global variance in the image. As a result, the values of local variance for one image cannot be directly compared with those from other images. Measurements of local variance derived from an image have meaning only when compared with other measurements derived from the same image degraded to different resolutions.

Results

Graphs of local variance as a function of resolution were calculated for images from three kinds of environments at two different resolutions. It must be remembered that the results are derived from a single image for each range of resolutions for each environment and slightly different results will be obtained from different images. While the shapes of the graphs may not change drastically, the specific resolutions associated with various portions of the graph will change depending on the size of the objects in particular scenes. The method of thinking about the interaction of environments and spatial resolutions is more important than the specific results presented from this limited sample of images.

For each type of environment there are two graphs of local variance as a function of resolution, one derived from an image with fine spatial resolution, and one from an image from the Thematic Mapper or the Thematic Mapper Simulator (TMS). The use of two images for each type of environment allows measurement of local variance over a wider range of spatial resolutions. As noted earlier, the image degradation process reduces the number of pixels in the image, limiting the range of resolutions covered by individual

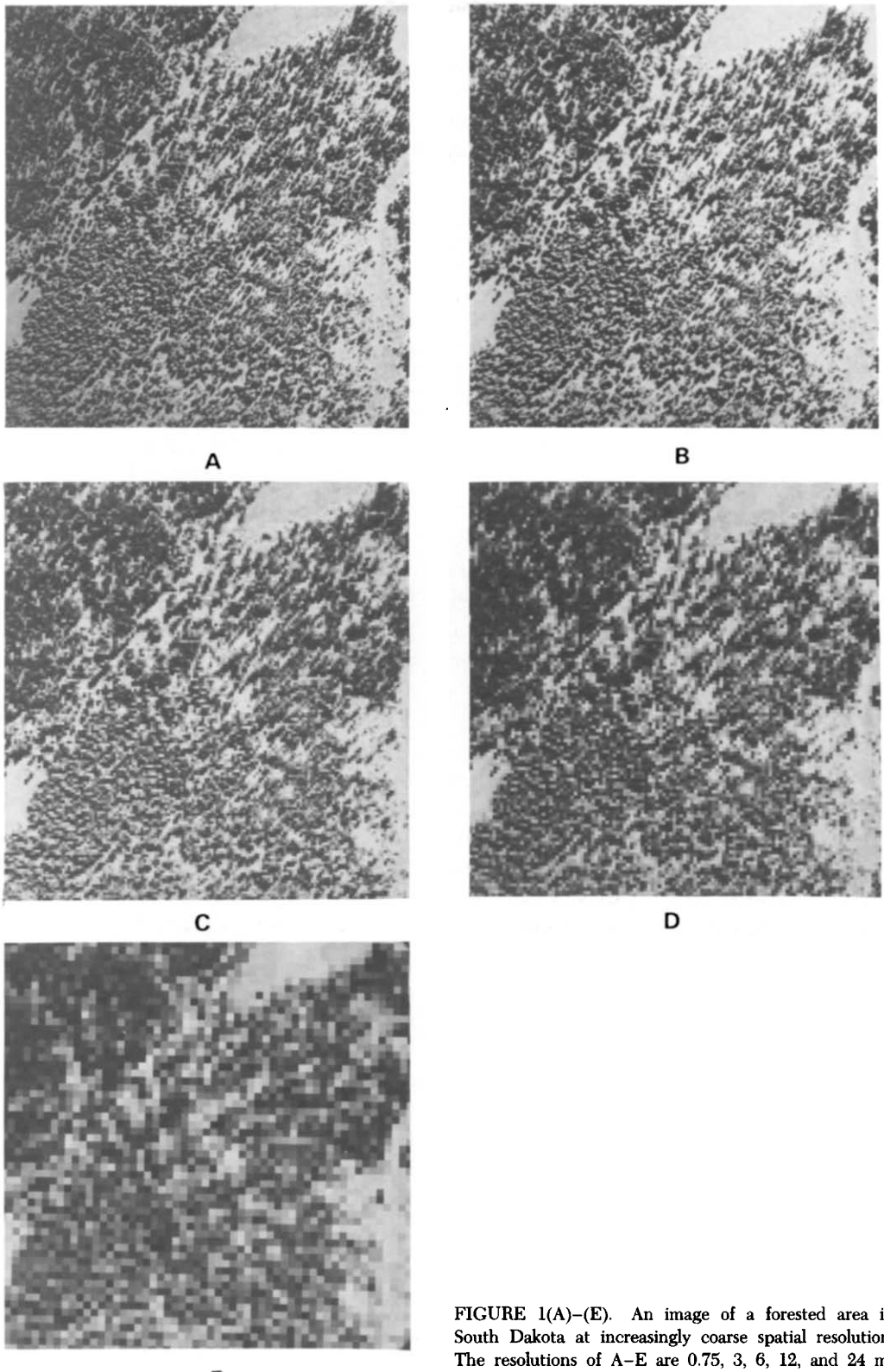


FIGURE 1(A)–(E). An image of a forested area in South Dakota at increasingly coarse spatial resolution. The resolutions of A–E are 0.75, 3, 6, 12, and 24 m, respectively.

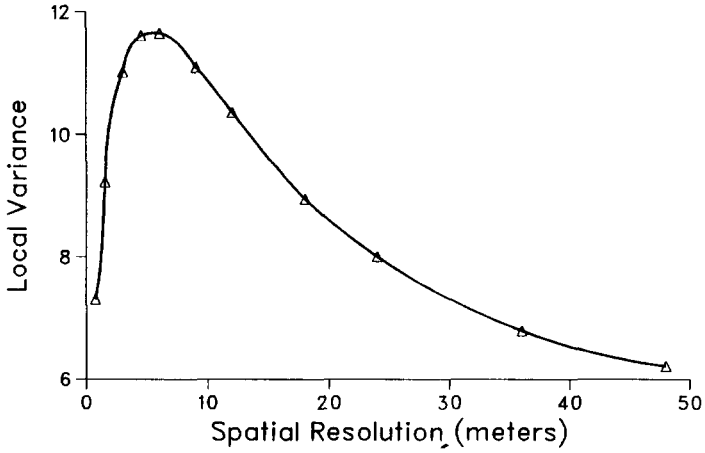


FIGURE 2. The graph of local variance as a function of resolution for the forest image shown in Fig. 1. The initial spatial resolution of the image was 0.75 m.

graphs. Because of the ad hoc nature of the measurement of local variance, these graphs cannot be directly combined. Instead, only their general shape for overlapping resolutions can be compared.

Forest images

To create a forest image with very fine spatial resolution, an 1:15,000 aerial photograph of a forested area in South Dakota was scanned using a microdensitometer [Fig. 1(A)]. The exact location of the area covered in South Dakota is unknown, but it serves as a good example of a simple forest scene composed of trees on a relatively smooth background. The spatial resolution is approximately 0.75 m, and a red filter was used in scanning the image. The graph of local variance as a function of resolution for this image is shown in Fig. 2. Local variance is low at the initial resolution of the image. Resolution cells are much smaller than trees, as tree crowns in this area are about 8 m in diameter. If a pixel falls on a tree, its immediate neighbors are also likely to be on a tree and have similar values. In this situation, the standard deviation of a 3×3 window is low. Naturally, some pixels will fall along the borders of trees and background, and as a result will have high local variance. However, the mean value of the local variances for the entire image is low.

As the size of individual resolution cells increases, the number of pixels comprising an individual tree decreases, and the likelihood that surrounding pixels will be similar decreases [Fig. 1(B)]. In this situation, local variance increases. This trend continues until a peak in local variance is observed at 6 m [Fig. 1(C)], when the pattern in the image becomes very mottled. As resolution increases past this peak, local variance decreases. This decrease is associated with individual pixels increasingly being characterized by a mix of both trees and background. As this mixing of elements occurs, all pixels look more similar and local variance continues to decrease [Fig. 1(D) and 1(E)].

A TMS image (Band 3) of an area in northern California near Mt. Shasta was used as an example of a forest environment at 30-m resolution (Fig. 3). The area is reasonably flat and is primarily eastside

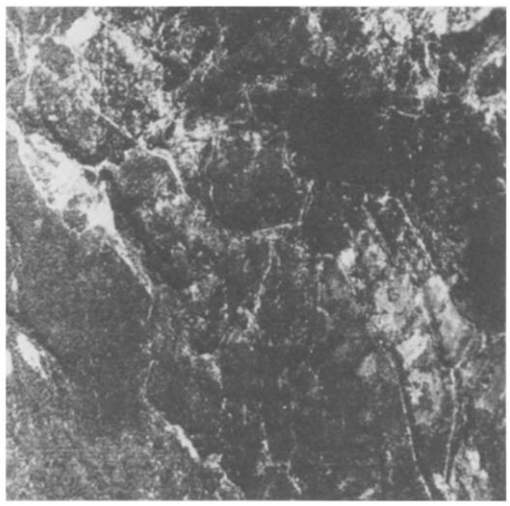


FIGURE 3. A Thematic Mapper Simulator image (Band 3) of a forested area in northern California at 30-m resolution.

pine, a vegetation association that runs along the eastern slopes of the Sierra Nevada and continues in extensive stands on many dry, flat areas of northeastern California. *Pinus Jeffreyi* and *P. ponderosa* are the dominant tree species in stands that tend to be sparsely stocked with a broken understory of shrubs and grasses.

The graph of local variance as a function of resolution for the TMS forest image is shown in Fig. 4. The graph begins with high local variance at 30-m resolution and declines as resolution-cell size increases. At 30-m resolution in a forest environment, the trees are considerably smaller than the resolution cells, and thus are not useful as elements in the scene model. Instead, areas within which the characteristics of the trees are similar, or stands, become the elements. Initially it was expected that a second peak in local variance would be found in forest environments that was related to the size of stands. One possible reason this peak does not occur is that the sizes of stands in this scene vary so widely that there is not one size dominant enough to cause a peak. It is also possible that the use of an image from another area might yield the hypothesized result, as this scene does not exhibit well-defined stands. In particular, the flat topography in the area may be adversely affecting the formation of distinct stands, as stand boundaries are often associated with topography in steep terrain.

Urban/suburban images

An image of a portion of Canoga Park, CA was used as an example of a resi-

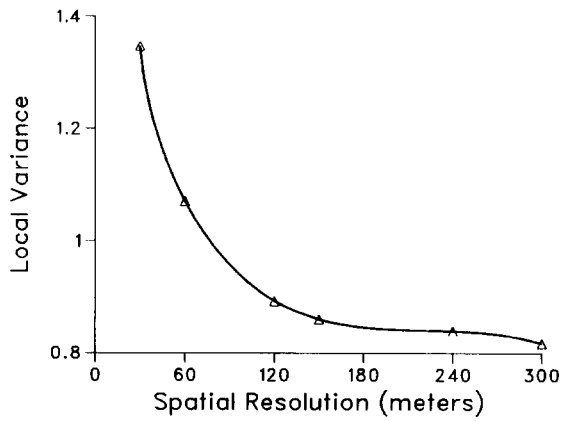


FIGURE 4. The graph of local variance as a function of resolution for the TMS image of a forest area shown in Fig. 3.



FIGURE 5. An image at 2.5-m resolution of a portion of a housing development in Canoga Park, CA.

dential environment (Fig. 5). The image is from the red portion of the spectrum and has approximately 2.5-m resolution. This scene is complex in nature, having several kinds of elements arranged in a mosaic. The most obvious elements are houses (or roofs from the aerial perspective), streets, and yards. The graph of local variance as a function of resolution (Fig. 6) begins with low local variance at the original resolution of the image, 2.5 m. At this resolution, pixels are smaller than the objects in the scene. Local variance increases until a peak is reached between 10 and 15 m, or four to six times the original resolution. Local variance declines over the remaining resolutions. The peak in the graph occurs at a resolution-cell size somewhat smaller than the size of the objects in the scene. Houses are generally 10–12 pixels wide, while the streets are 6–8 pixels and the spaces between houses and between houses and streets are 3–10 pixels.

A TM image of Washington DC was used as an example of an urban/suburban environment. The image is the red band (Band 3, 0.63–0.69 μm) on 2 November 1982 (Fig. 7). The graph of local variance as a function of resolution (Fig. 8) starts with high local variance at the initial resolution (30 m) and stays high after the image has been degraded once to 60 m. After this initial flat region, local variance declines. The lack of a well-developed peak in local variance indicates there is not a distinct group of objects of a specific size that dominate the scene within the range of resolutions covered in the graph.

The interpretation of the flat portion at the beginning of the graph is difficult. It is possible that it corresponds to a peak in

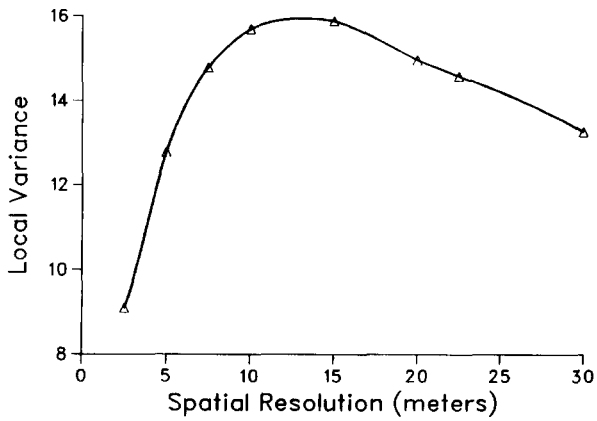


FIGURE 6. The graph of local variance as a function of resolution for the image of Canoga Park, shown in Fig. 5.



FIGURE 7. A TM image (Band 3) of Washington, DC.

local variance, the entire shape not being revealed due to the limitation of the initial spatial resolution of the data. The results from the Canoga Park image indicate local variance is in a well-developed decline by 30 m. However, the sizes of objects may be sufficiently larger in the Washington, DC, scene that the peak in the local variance is shifted to coarser resolutions.

The behavior of local variance in the region between the peak in the graph for Canoga Park (10–15 m) and the flat start of the graph for Washington, DC is important because it encompasses significant resolutions from the perspective of new and planned space-borne sensors. In urban/suburban environments, it is our expectation that the location of a peak in local variance will shift on the basis of individual scenes. At one extreme will be peaks at resolutions as low as 8–10 m in scenes dominated by residential developments of single family dwellings. As the scene changes to a more urban character, the sizes of objects in the scene will increase and the peak will shift to coarser resolutions, possibly 20–30 m. In addition, a cultural influence on land parcel size has been documented that will affect the location of the peak (Jensen, 1983). In the long run, the exact location of the peak is not the main issue. The main conclusion to be drawn from these graphs is that the spatial resolutions of some of the new sensors like TM and SPOT are generally characterized by high local variance in urban and suburban scenes.

Agricultural images

To produce an image of an agricultural environment at very fine resolution (0.15 m), a 1:3000 color-infrared aerial

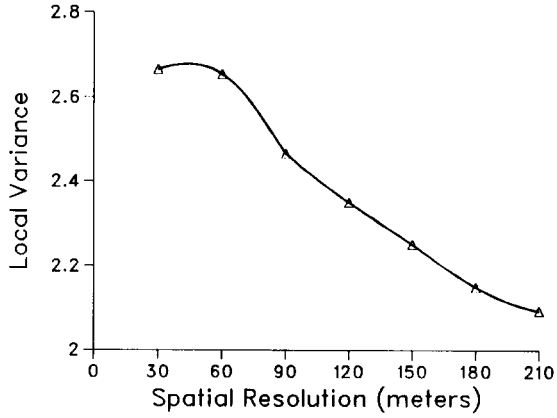


FIGURE 8. The graph of local variance as a function of resolution for the TM image of Washington DC shown in Fig. 7.

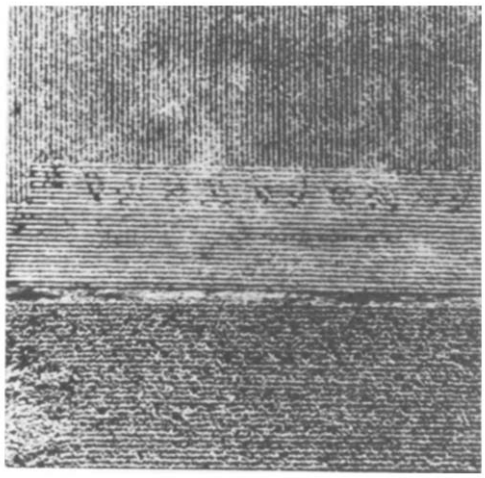


FIGURE 9. An image of an agricultural area at 0.15-m resolution created by scanning an aerial photograph.

photograph of agricultural fields in Oklahoma was scanned using a microdensitometer (Fig. 9). The crops, corn and soybeans, exhibit a distinct row structure and are near maturity as the canopy is almost closed. This image is relatively simple in structure, with crop rows, shadows, and an almost entirely obscured soil background as the only elements. The graph of local variance as a function of resolution (Fig. 10) is similar in shape to the graph from the Washington, DC image (Fig. 8) but covers dramatically different resolutions. At the initial resolution of the image and after one degradation, local variance is high. Local variance then decreases rapidly to a flat area of low local variance. It was originally hoped that the initial resolution of the data would be fine enough to show a definite peak in the graph associated with the size of crop rows since they are about 5 pixels apart in the image. However, there is considerable variation over those 5 pixels, from illuminated canopy to shadowed canopy to the gap between rows. As a result, local variance is high at the initial resolution.

A TM Band 3 image from 22 August 1982 was obtained from an area near Dyersburg, TN to serve as an example of an agricultural environment (Fig. 11). The area looks like a patchwork of homogeneous blocks. At this resolution the elements are now entire fields rather than the crop rows that comprise the fields. The image is large, and therefore the graph of local variance as a function of resolution covers a wide range of resolutions (Fig. 12). At the initial resolution (30 m) local variance is relatively low.

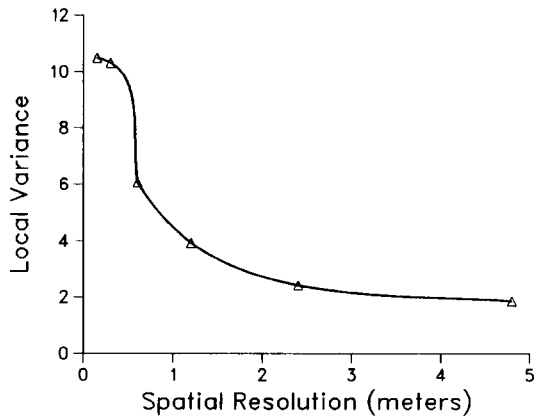


FIGURE 10. The graph of local variance as a function of resolution for the agricultural image shown in Fig. 9.



FIGURE 11. A TM image (Band 3) of an agricultural area near Dyersburg, TN.

While there is a large gap between the two graphs, local variance is likely to remain low, between 5- and 30-m resolution in agricultural areas as there are not intermediate size elements between crop rows and fields. After 30 m, local variance rises to a peak at 240 m, or eight times the initial resolution. After the peak in local variance, there is a general decline over the remaining resolutions in the graph.

The agricultural graphs of local variance as a function of resolution indicate two distinct scales of high variance, one related to the size of individual crop rows, and one related to the size of fields. The first peak occurs at resolutions obtainable only from airborne scanner systems. However, the important point that multiple scales of variation exist in an environment is illustrated.

Discussion

One methodological issue resulting from the observed graphs of local variance as a function of resolution concerns the relationship between the size of the objects in the scene and the spatial resolution of peak local variance. It was initially hypothesized that the peak would occur when the size of the resolution cells matched the size of the objects. However, in each of the graphs with a well-developed peak in local variance the peak occurs at a resolution-cell size somewhat smaller than the size of the objects in the scene. In the South Dakota forest image, the peak occurs at 6 m for trees approximately 8 m in diameter. The peak in local variance occurs at 240 m for agricultural fields generally 420 m on a side

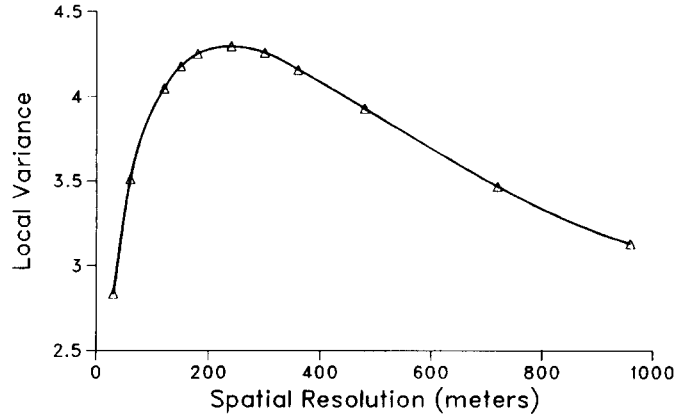


FIGURE 12. The graph of local variance as a function of resolution for the TM image of an agricultural area shown in Fig. 11.

in the area near Dyersburg, TN. It is more difficult to determine the average size of objects in the Canoga Park scene, but 20–25 m is a good estimate. In this case, the peak in local variance is in the range of 10–15 m.

To investigate this situation, graphs of local variance as a function of resolution were calculated for two simulated images of known characteristics. These images were simulated using a simplified version of a program developed for modeling bidirectional reflectance of coniferous forest canopies (Li and Strahler, 1985). One image is simply black discs 7 pixels in diameter randomly located on a white background (Fig. 13). The peak in the graph of local variance as a function of resolution for this image occurs at a resolution between 5 and 6 times the initial resolution for discs 7 pixels in diameter (Fig. 14).

The second simulated image is of a forested scene. Trees are modeled as cones and are randomly located on a white background. The cones are illuminated from a solar zenith angle of 20°, resulting in four kinds of pixels in the image: illuminated tree canopy, illuminated background, shadowed tree canopy, and shadowed background (Fig. 15). The peak in the graph of local variance as a function of resolution for the simulated forest image (Fig. 16) occurs at 6 pixels for objects 7 pixels in diameter along one axis and 11 pixels along the other axis. As with the observed images, the peak in local variance for both simulated images occurs before the size of the objects. For all the observed images with definite peaks in local variance and the simulated images, the peak in local variance occurs in the range of 1/2 to 3/4 of the size of the objects in the scene. While this contradicts the portion of the initial hypothesis that local variance would peak at the size of the objects, at the same time it supports the basic underlying idea that local variance in images is related to the size of the objects in the scene and spatial resolution.

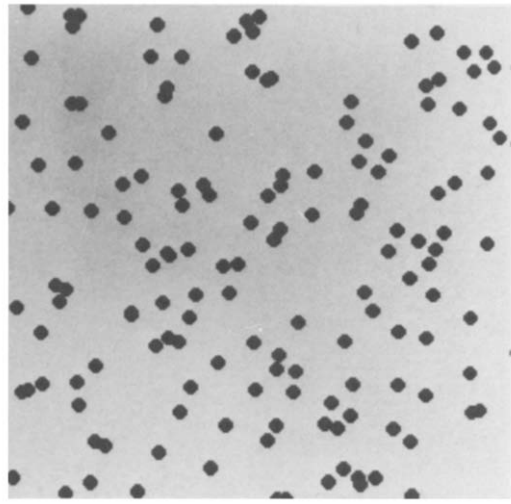


FIGURE 13. A portion of a simulated image of black discs on a white background. The discs are randomly located and are 7 pixels in diameter.

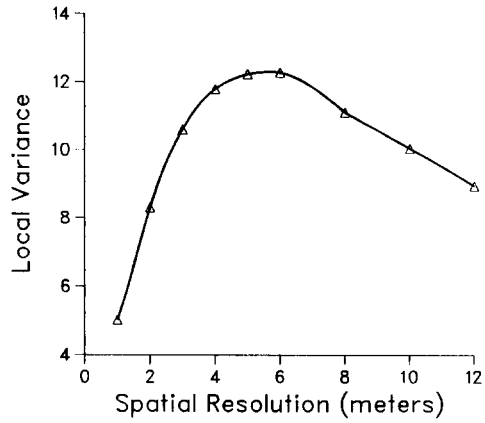


FIGURE 14. The graph of local variance as a function of resolution for the simulated disc image shown in Fig. 13.

Examination of the changes in both the simulated forest image and the local variance image derived from it as the image

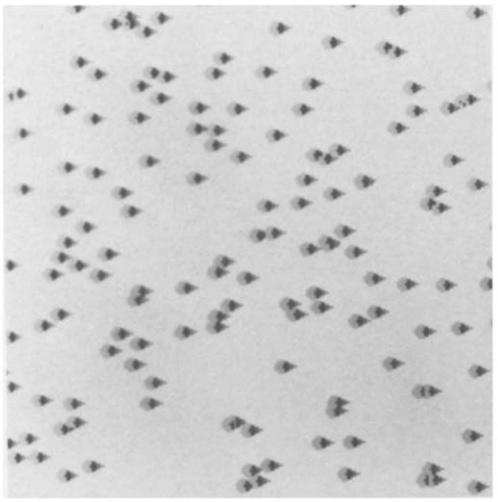


FIGURE 15. A portion of a simulated forest image in which trees are modeled as inverted cones. The trees are illuminated from a solar zenith angle of 20° from the east (right side of the page).

is degraded to coarser resolutions helps explain the reason local variance peaks before the size of the objects. To display this process, a series of pictures with portions of both the simulated image and its associated local-variance image are placed side by side in Figures 17(A)–(D). In interpreting these images, it should be remembered that the area covered by each successive picture increases. The first picture [Fig. 17(A)] shows the simulated image at its original resolution of 1 m. In the local-variance image, high local variance (light tones) occurs primarily around the perimeter of the trees and their shadows, much like an edge detector. The area inside the perimeter still has relatively low local variance and the area between trees is black (zero local variance), as the background has the same value in all locations.

Figure 17(B) shows the results after the image has been degraded to 3 m. At this resolution the trees cannot be distinguished from their shadows and begin to appear out of focus. The dark areas inside trees in the local variance image have disappeared because the size of the trees in numbers of pixels has decreased. Similarly, the number of pixels between trees has decreased and local variance is influenced by effects from neighboring trees. Comparison with the first local variance image [Fig. 17(A)] shows a larger portion of the area covered by bright values, indicating higher overall local variance.

The resolution of peak local variance (6 m) is shown in Fig. 17(C). At this

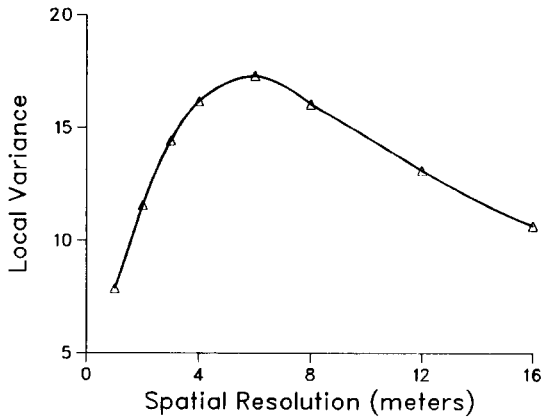


FIGURE 16. The graph of local variance as a function of resolution for the simulated forest image shown in Fig. 15.

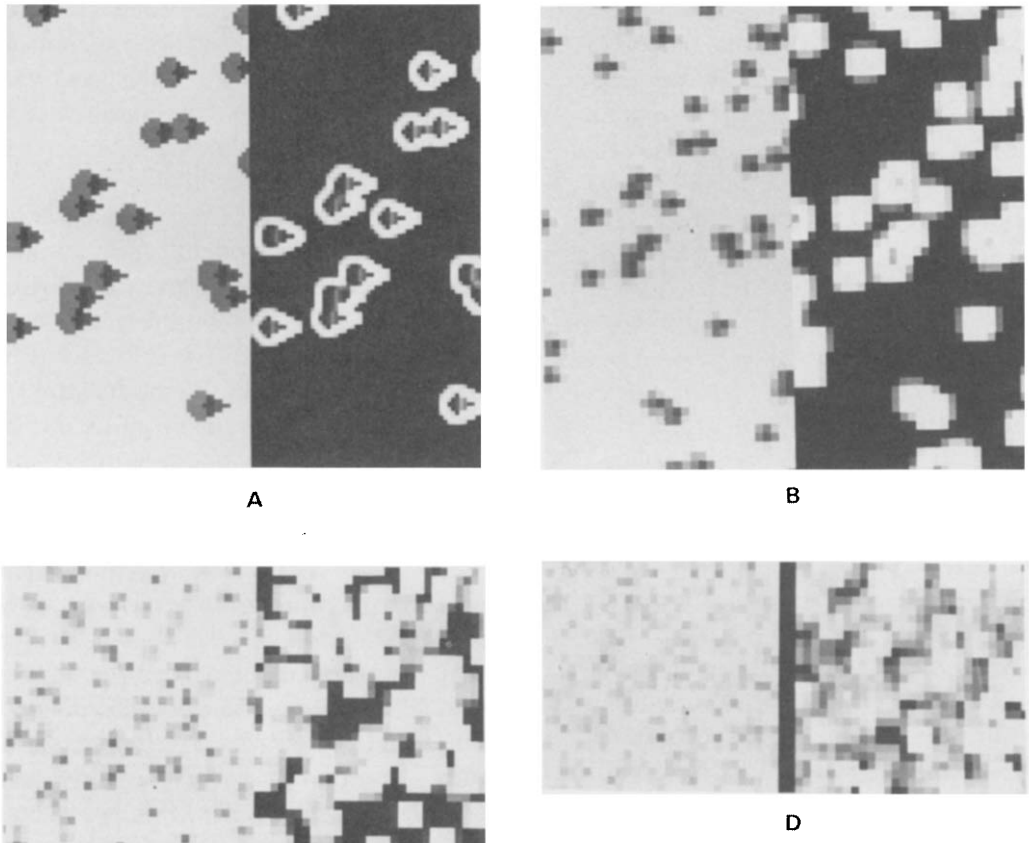


FIGURE 17(A)–(D). The effect of coarser resolution on the simulated forest image and derived local-variance images. Each figure shows the derived local-variance image next to the simulated forest image. The resolution for A–D are 1, 3, 6, and 9 m, respectively.

resolution, trees have become very small, and a large area of the local variance image is bright, indicating high local variance. An interesting characteristic of this image is that there are a considerable number of pixels with intermediate values in the local variance image. In the previous local variance images, pixels were generally near edges and very bright, or in homogeneous areas and very dark. These intermediate local-variance values are the result of the effect of the degradation of resolution on the appearance of trees.

At a resolution of 9 m, local variance has declined [Fig. 17(D)]. The local variance image has begun to look like a continuous tone image, quite different from the edge detector in Fig. 17(A). While a greater proportion of the local variance image has values other than black, the mean value of the image is lower. This observation is the key to understanding the reason that local variance peaks be-

fore the size of the elements. As the imagery is degraded, the model for the appearance of a tree is different from that originally expected. As the resolution cells become larger, trees tend to look increasingly out of focus, with many pixels being composed of a mixture of both dark tree or shadow and light background. Thus, as the size of a tree is approached, instead of having alternating light and dark pixels for tree or background, there are several intermediate tone pixels. The effect of any given tree is spread through several pixels. This effect can be seen in Figs. 17(C) and 17(D). The effect of numerous intermediate-tone pixels is the production of only a few high local-variance values. The contrast between pixels in the image is generally not large enough for high local-variance values.

When viewed from this perspective, the result that local variance peaks before the size of the objects makes sense. The sampling theorem states that a resolution-cell size less than half the size of the objects in the scene would be necessary to assure full contrast in the measurements in the image. In addition, as resolution-cell size increases, there is a decrease in the area covered by low local-variance values. The combined result of these effects is peaks in local variance about 1/2–3/4 the size of the objects in the scene in both the simulated and observed images.

The finding that peaks in local variance do not occur exactly at the size of the objects in the scene is not as important as the finding that the peak is directly related to the size of the objects. The general relationship between local variance and spatial resolution is of primary importance. Knowing whether local variance is generally high or low for each resolution is the primary information desired.

The effect of scale on image classification

The graphs of local variance as a function of resolution provide a method of measuring the interaction of environments and the spatial resolution of sensors. While this relationship is intrinsically interesting, it has implications for other issues in remote sensing. In particular, the graphs of local variance as a function of resolution provide useful insights for a lingering question in remote sensing; the effect of the interaction of spatial resolution and environments on image classification.

A comprehensive study of the effect of spatial resolution on classification accuracy was undertaken by Markham and Townshend (1981), and their conclusions represent the culmination of the results of many earlier studies. They concluded that observed classification accuracies were the result of a tradeoff of two factors. The first factor is the influence of boundary pixels on classification results. As spatial resolution becomes finer, the proportion of pixels falling on the boundary of objects in the scene will decrease. Boundary pixels have a mix of elements, and reducing the number of mixed pixels reduces confusion in the classification process, resulting in higher classification accuracy.

The increased spectral variance of land-cover types associated with finer spatial resolution is the second factor identified as influencing classification accuracy. Within-class variance decreases the spectral separability of classes and results in lower classification accuracy.

This factor is often referred to as "scene noise." The net effect of finer spatial resolution is the result of the combination of these two opposing factors which varies as a function of environment.

The graphs of local variance as a function of resolution are particularly pertinent to the "scene noise" factor. While the description of the process is lucid, the term "scene noise" is ill advised, as it has a distinct negative connotation. It implies that the within-class variance is either an artifact or unwanted. One of the main points illustrated by the graphs of local variance as a function of resolution is that it is not an artifact, but an understandable, even predictable effect of finer resolution in scenes best characterized by a complex and nested scene model. At finer resolutions, the elements at the lower level in the nested model increasingly influence the spectral values associated with the classes of the higher level. Because of the complex nature of the scenes, the spectral variability of a class will undoubtedly increase under these conditions.

To understand the perspective that increased within-class variance is unwanted is considerably more complicated. Essentially, an understanding of the nature of image classification and the underlying assumptions of the classification methods is required. However, the significance of the question warrants the discussion.

It is assumed in image classification that the ground scene can be partitioned into categories or classes. These classes in most environments are not pure, or spatially homogeneous. Thus, scene models for classification usually exhibit a nested structure. At the higher level, there are the classes that are to be differentiated, and at a lower lever there are usually several (or often times many) different kinds of elements that when combined characterize the class. Different classes in a scene may vary from the extreme case of having completely different sets of component elements to sharing exact sets.

From a practical point of view, while the classes can be abstractly defined and thus theoretically differentiated, in the real world there tends to be a continuum of variation between classes. The difference between classes can be as subtle as varying proportions of the same component elements. Thus, it is not always possible to determine the class of a given point; instead there is a certain amount of spatial integration involved to take into account the context of the point. For example, consider a natural environment in the transition area between the high desert and the adjacent forest. The two classes can be described as quite different, but grade together by subtle differences in the elements comprising the two classes. In the high desert environment there is a general cover of brush with scattered trees. In the adjacent forest there is a broken canopy of trees with a brush understory. With all possible combinations occurring between, it becomes difficult to determine where the transition occurs. For any given point on the landscape it may be impossible to determine which environment it most closely approximates without looking at the surrounding area.

With the concept of a complex, nested scene model in mind, it is interesting to discuss different classification methods, particularly with respect to their assumptions concerning the spatial structure of images. Combining this discussion with the graphs of local variance as a function of resolution, it is possible to determine

the combinations of resolutions and environments most appropriate for each classification method.

Spectral classifiers. Most classification approaches commonly used in remote sensing rely solely on spectral data to differentiate classes. Each pixel is processed independent of its neighbors or location in the image. Individual measurements are assumed to be samples from a larger element. For a complex, nested scene model, the implicit assumption is that the resolution cells are large enough that sufficient spatial integration of the elements comprising classes occurs that the individual measurements can be properly classified on the basis of the mix within the pixel.

However, it is not common in a complex, nested scene model that this assumption of each pixel containing a representative mix of component elements remains valid. This assumption begins to break down as elements comprising the classes in the scene become larger relative to the resolution cell. High local variance in an image is indicative of this situation. As this occurs, classification is dominated by identification of elements in the lower level of the nested scene model. This produces confusion in scenes where different classes share common elements.

These classifiers are most appropriate for combinations of environments and spatial resolutions with low local variance. They were originally developed for use with Landsat MSS data at 80-m resolution, which has relatively low local variance for all environments tested, helping explain their popularity and widespread use. They have been used in most of the empirical tests of the effect of changing spatial resolution on classification accuracy. The results of these tests are consistent with the results that would be expected from combining a classification method that assumes low local variance and the results shown in the graphs of local variance as a function of resolution.

With one exception, the results of experiments in forest environments indicate that classification accuracies decrease as spatial resolution becomes finer than 60–80 m (Latty and Hoffer, 1980; Sadowski et al., 1977; and Kan et al., 1975). In the exception, Teillet et al. (1981) reported no significant change in classification accuracy at 30 and 50 m. As Figures 2 and 4 indicate, there is a general increase in local variance as spatial resolution decreases from the 60–80 m range. Thus, the implicit assumption of low local variance of the classifiers becomes less valid, explaining the declining accuracies. Interestingly, at very fine spatial resolutions (below 3 or 4 m), local variance is low, and the assumptions of the spectral classifiers are again valid. However, these classifiers would produce results about individual trees rather than stands of trees at these resolutions.

Similar results to the forestry experience have been found in tests of finer spatial resolution in urban/suburban environments. Comparing the results of MSS and TM Data using spectral classifiers, Toll (1983) found that land-cover discrimination decreased using TMS data as a result of increased within-class variability. Clark and Bryant (1977) also found classification accuracies decreased at finer spatial resolutions in an urban/suburban environment. Figures 6 and 8 indicate local variance increases at finer spatial resolutions which violates the assumptions of the spectral classifiers.

In agricultural environments, Fig. 12 indicates that the opposite effect on clas-

sification accuracy would be expected for resolutions finer than 80 m because local variance decreases and the assumptions of the spectral classifiers become more valid. The problems of mixed pixels encountered during the LACIE and AgRISTARS programs should be reduced using TM data (30 m). The results of Bizzell and Prior (1983) support this expectation. However, Ahern et al. (1980) report lower classification accuracy for 30 m data than 80 m data. This decrease is attributed to the increase in spectral variability within fields, which in this case is believed to outweigh the increased boundary problems at 80 m. The size of fields and the spectral separability of the crop types will influence this tradeoff.

Mixture models. Another group of classification methods are designed for a different situation and use proportion estimation or mixture models. In these models, the proportion of several elements is estimated for each pixel (Marsh et al., 1980; Horowitz et al., 1975; Adams et al., 1982). Mixture models concentrate on the elements comprising the scene that are smaller than the resolution cells. Recovery of information about higher levels in a nested scene model is not possible.

Mixture models are most appropriate under conditions of high local variance where the contrast between measurements is maximized, but their formulation does not exclude their use under low local-variance conditions. The versatility of mixture models should result in their successful application over a wide range of resolutions and environments. Because of their formulation for high variance conditions, applications in forested and urban/suburban environments with data from sensors like TM and SPOT would be logical. However, the mixture models provide information about the elements in the lower level of the scene model, and as such are not a substitute for the spectral classifiers at these resolutions. In forest environments, the mixture models would estimate the proportion of pixels covered by elements such as trees and background, rather than classifying pixels according to the characteristics of the trees in the pixel. Similarly, in urban/suburban environments they would provide information about the elements comprising the scene, such as houses, lawns, roads, etc., not classify pixels according to general land use or land cover.

Spatial and spectral classifiers. Another group of classification approaches assume conditions of high spatial autocorrelation in the image. The assumed scene model consists of spatially homogeneous objects that are larger than the resolution cells in the image, resulting in low local variance in the image data. The basic strategy of these classifiers is to exploit the spatial correlation in the data by combining neighboring pixels into multipixel units as part of the classification process. There are a variety of algorithms used to accomplish this task, but they are all based on the idea that classification is enhanced by using information from surrounding pixels (Kauth et al., 1977; Landgrebe, 1980; Bryant, 1979). For these approaches, the spatial integration often required for determination of class membership is not assumed to occur within individual pixels.

The conditions of low local variance assumed by these classifiers limit the combinations of environments and spatial resolutions in which they are appropriately used. Agricultural scenes most closely approximate the assumed scene model, with resolutions of low local variance between the effects of crop rows

and fields being most appropriate for their use. Possibly in forests characterized by distinct stands these classifiers might be successfully employed with imagery coarse enough to be unaffected by individual trees.

Texture. Another group of classification approaches that incorporate information from surrounding pixels use texture measures. To use texture measures in image classification, the value of one or more texture measure is calculated for each pixel in the image, and these values are then used to determine the class of pixels. Often for image classification, texture measures are used in conjunction with spectral data (Hsu, 1978; Jensen, 1979; Woodcock, et al., 1980; Shih and Schowengerdt, 1983; Frank, 1984). Haralick (1979) provides an excellent review of the texture measures used in remote sensing as well as other applications in image processing. The use of image texture explicitly implies that the resolution cells are smaller than the elements in the scene model, because numerous measurements are required for each element or class in order to allow the characteristic spatial texture to occur. However, the difference between the texture approaches and the last group concerns the nature of the assumed scene model. The use of texture implies that the elements in the scene model are not spatially homogeneous. Essentially it is the inhomogeneity of the elements that produce the textures characteristic of different classes.

Because of their reliance on spatial variation to differentiate classes, texture methods are most appropriate under conditions of high local variance. Thus, they are most appropriate at spatial resolutions opposite of the last group of classifiers. This indicates that the use of texture measures in image classification for urban/suburban and forest environments with TM and SPOT would be logical. This idea was expressed by both the Botanical Sciences and Land Use/Land Cover Teams of the Multispectral Imaging Science Working Group (Cox, 1982).

Other attempts to incorporate spatial considerations in image analysis or classification include contextual classifiers (Wharton, 1982; Tilton, 1983; Yu and Fu, 1983), expert systems and artificial intelligence. These approaches attempt to incorporate in automated methods information sources such as shape, site, association, and context, which are routinely used by photointerpreters (Estes, 1977). One common characteristic of these methods is that they are most effective under conditions of high local variance. Thus, their significance should increase with the use of TM, SPOT, and other fine-resolution sensors.

Conclusions

A new method has been presented that measures the spatial structure of images as a function of spatial resolution. These graphs have illustrated several points about the local variance in images:

The local variance in images is related to the relationship between the size of the objects in the scene and the spatial resolution of the sensor.

The spatial resolutions of high local variance change as a function of environment.

Multiple scales of variation in an environment will produce multiple ranges of spatial resolution with high local variance.

The concept of a scene model is particularly important to the interpretation of

the graphs of local variance as a function of resolution. The definition of discrete scene-model elements is required for understanding and interpreting the locations of peaks in local variance. Also, the concept of nested scene models is helpful when the graphs of local variance as a function of resolution cover wide ranges of spatial resolutions and multiple scales of variation in images.

The graphs of local variance as a function of resolution for the urban/suburban, agricultural, and forested scenes generally support and help explain the results found by individual investigators in studies of the effect of spatial resolution on classification accuracy. An explicit understanding of the assumptions of spatial structure made by classifiers is critical in understanding the results of these tests. The domination of spectral classifiers has led to a limited view of the nature of scenes. Rather than evaluating classifiers on the basis of the validity of their assumptions for a particular scene and spatial resolution, spatial resolutions are being evaluated on the basis of a particular classifier.

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References

Adams, J. D., Smith, M., and Adams, J. R. (1982), Use of laboratory spectra for determining vegetation assemblages in Landsat images, *Int. Symp. on Remote Sensing of Environment, Second Thematic Conference, Remote Sensing for Geologic Exploration*, Fort Worth, TX, pp. 757–771.

Ahern, F. J., Brown, R. J., Goodenough, D. G., and Thomson, K. P. B. (1980), A simulation of Thematic Mapper performance in an agricultural application, *Proc. of the Sixth Canadian Symp. on Remote Sensing*, Halifax, NS, pp. 585–596.

Bizzell, R. M., and Prior, H. L. (1983), Thematic Mapper quality and performance assessment in renewable resources/agriculture/remote sensing, *Proc. of the Landsat-4 Science Characterization Early Results Symposium*, Greenbelt, MD, Vol. IV, pp. 299–312.

Bryant, J. (1979), On the clustering of multidimensional pictorial data, *Pattern Recognition* 11:115–125.

Clark, J., and Bryant, N. A. (1977), Landsat-D Thematic Mapper Simulation using aircraft multispectral data. *Proc. of the 11th Int. Symp. on Remote Sensing of Environment*, Ann Arbor, MI, pp. 483–491.

Cliff, A. D., and Ord, J. K. (1981), *Spatial Processes: Models and Applications*, Pion, London.

Cox, S. D. (Ed.) (1982), The Multispectral Imaging Science Working Group: Final Report, NASA Conference Publication 2260.

Craig, R. G., and Labovitz, M. L. (1980), Sources of variation in Landsat autocorrelation, *Proc. of the 14th Int. Symp. on Remote Sensing of Environment*, San Jose, Costa Rica, pp. 1755–1767.

Engel, J. L., and Weinstein, D. (1983), The Thematic Mapper—an overview, *IEEE Trans. Geosci. Remote Sens.* 21(3): 258–265.

Estes, J. (1977), A perspective on the state of the art of photographic interpretation, *Proc. of the 11th Int. Symp. on Remote Sensing of Environment*, Ann Arbor, MI, pp. 161–177.

Frank, T. D. (1984), The effect of change in vegetation cover and erosion patterns on albedo and texture of Landsat images in a semiarid environment, *Ann. Assoc. Am. Geographers* 74(3):393–407.

Greig-Smith, P. (1964), *Quantitative Plant Ecology*, Butterworths, London.

Haralick, R. M. (1979), Statistical and structural approaches to image texture, *Proc. IEEE* 67(5):768–804.

Horowitz, H. M., Lewis, J. T., and Pentland, A. P. (1975), Estimating the proportions of objects from multispectral scanner data, Environmental Research Institute of Michigan, Final Report N. 109600-13-f, 117 pp.

Hsu, S. (1978), Texture-tone analysis for automated land-use mapping. *Photogramm. Eng. Remote Sens.* 11:1393–1404.

Ishizawa, Y. (1981), The Japanese MOS and LOS program, *Proc. of the 15th Int. Symp. on Remote Sensing of Environment*, Ann Arbor, MI.

Jensen, J. (1979), Spectral and textural features to classify elusive land cover at the urban fringe, *Professional Geographer*, 4:400–409.

Jensen, J. R. (1983), In *The Manual of Remote Sensing*, (R. N. Colwell, Ed.), American Society For Photogrammetry and Remote Sensing, Falls Church, VA, Chap. 31, pp. 1571–1666.

Kan, E. P., Ball, D. P., Basu, J. P., and Smelser, R. L. (1975), Data resolution versus forestry classification accuracy, *Symposium on Machine Processing of Remotely Sensed Data*, Purdue University, West Laffayette, IN, pp. 1B-24, 1B-36.

Kauth, R. J., Pentland, A. P., and Thomas, G. S. (1977), BLOB: An unsupervised clustering approach to spatial preprocessing of MSS imagery, *Proc. 11th Int. Symp. Remote Sens. Environ.* 2:1309–1317.

Labovitz, M. L., Toll, D. L., and Kennard, R. E. (1980), Preliminary evidence for the influence of physiography and scale upon the autocorrelation function of remotely sensed data, NASA TM 82064, Goddard Space Flight Center, Greenbelt, MD.

Landgrebe, D. A. (1980), The development of a spectral-spatial classifier for earth observational data, *Pattern Recognition* 12:165–175.

Latty, R. S., and Hoffer, R. M. (1980), Computer-based classification accuracy due to the spatial resolution using per-point versus per-field classification techniques, *Machine Processing of Remotely Sensed Data Symposium*, West Lafayette, IN.

Li, X., and Strahler, A. H. (1985), Geometric-optical modeling of a conifer forest canopy, *IEEE Trans. Geosci. Remote Sens.* GE-23(5):705–721.

Markham, B. L., and Townshend, J. R. G. (1981), Land cover classification accuracy as a function of sensor spatial resolution, *Proc. of the 15th Int. Symp. on Remote Sensing of Environment*, Ann Arbor, MI, pp. 1075–1090.

Marsh, S. E., Switzer, P., Kowalik, W. S., and Lyon, R. J. P. (1980), Resolving the percentage of component terrains within single resolution elements, *Photogramm. Eng. Remote Sens.* 46(8):1079–1086.

Midan, J. P., Reulet, J. F., Giraudbit, J. N., and Bodin, P. (1982), The SPOT HRV Instrument, *1982 International Geosciences and Remote Sensing Symposium*, Munich, West Germany.

Mollering, H., and Tobler, W. R. (1972), Geographical variances, *Geogr. Anal.* 4:35–50.

Rayner, J. N., (1971), *An Introduction to Spectral Analysis*, Pion, London.

Sadowski, F. G., Malila, J. E., Sarno, J. E., and Nalepka, R. F. (1977), The influence of multispectral scanner spatial resolution on forest feature classification, *Proc. of the 11th Int. Symp. on Remote Sensing of Environment*, Ann Arbor, MI, pp. 1279–1288.

Shih, E., and Schowengerdt, R. (1983), Classification of arid geomorphic surfaces using Landsat spectral and textural features, *Photogramm. Eng. Remote Sens.* 3:337–347.

Simonett, D. S., and Coiner, J. C. (1971), Susceptibility of environments to low resolution imaging for land-use mapping, *Proc.*

of the 7th Int. Symp. on Remote Sensing of Environment, Ann Arbor, MI, pp. 373–394.

Strahler, A. H., Woodcock, C. E., and Smith, J. A. (1986), On the nature of models in remote sensing, *Remote Sens. Environ.* 20:121–139.

Teillet, P. M., Guindon, B., and Goodenough, D. G. (1981), Forest classification using simulated Landsat-D Thematic Mapper data, *Can. J. Remote Sens.* 7(1):51–60.

Tilton, J. C. (1983), Contextual classification of multispectral image data using compound decision theory, *Proc. of the 17th Int. Symp. on Remote Sensing of Environment*, Ann Arbor, MI, pp. 863–871.

Toll, D. L. (1983), Preliminary study of information extraction of Landsat TM data for a suburban/regional test site, *Proc. of the Landsat-4 Science Characterization Early Results Symposium*, Goddard Space Flight Center, Greenbelt, MD, Vol. IV, pp. 387–402.

Townshend, J. R. G., (1980), The spatial resolving power of earth resources satellites: a review, NASA Goodard Space Flight Center TM 82020, Greenbelt, MD.

Tucker, C. J., Townshend, J. R. G. and Goff, T. E. (1985), African landcover classification using satellite data, *Science* 227(4685):369–375.

Tukey, J. W. (1961), Discussion, emphasizing the connection between analysis of variance and spectrum analysis, *Technometrics* 3:191–219.

Wharton, S. W. (1982), A contextual classification method for recognizing land use patterns in high resolution remotely sensed data, *Pattern Recognition* 15(4):317–324.

Woodcock, C. E., and Strahler, A. H. (1985), Relating ground scenes to spatial variation in images, *Proc. Third NASA Conference on Mathematical Pattern Recognition and Image Analysis*, Texas A&M University, College Station, TX, pp. 393–449.

Woodcock, C. E., Strahler, A. H., and Logan, T. L. (1980), Stratification of forest vegetation for timber inventory using Landsat and collateral data, *Proc. of the 14th Int. Symp. on Remote Sensing of Environment*, San Jose, Costa Rica, pp. 1769–1787.

Yu, T. S., and Fu, K. S. (1983), Recursive contextual classification using a spatial stochastic model, *Pattern Recognition* 16(1):89–108.

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